

● DAY-2 OFFLINE MODIFICATION

Theme: Predictive Safety & Decision-Grade Emergency Intelligence

🕒 Duration: 2 Hours

◆ Day-2 Context

Real public safety systems are **not reactive tools**.

Their true value lies in **anticipating escalation** and **guiding response actions** under uncertainty.

A system that only says “*something is wrong*” is insufficient.

A real system must answer:

- **What is likely to happen next?**
 - **How urgent is the situation becoming?**
 - **What action should be taken immediately?**
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🎯 Day-2 Objective

Extend your Day-1 **Risk Reasoning System** with **predictive or decision-support intelligence** that:

- Looks beyond the current frame
- Uses recent temporal patterns
- Outputs **actionable, explainable decisions**

Teams must implement **ONE** of the following **logic-centric extensions**.

🧠 DAY-2 MODIFICATION OPTIONS (CHOOSE ONE)

📌 OPTION 1: Short-Term Crowd Behavior Prediction

(Predictive Reasoning)

Problem

Authorities need early warnings **before** a situation becomes critical.

Required Logic

Your system must predict **near-future crowd behavior (5–10 seconds)** using **recent motion history**.

Predictions may include:

- Likely direction of dominant movement
- Probability of density increase or dispersal
- Likelihood of escalation from MEDIUM → HIGH risk

✓ Prediction may be probabilistic

✓ Accuracy is NOT the focus — reasoning is

Mandatory Output (Text / Log Based)

Time: 18:43:10

Prediction Horizon: 8 seconds

Predicted Behavior:

- Movement likely to intensify toward Zone C
- Density expected to increase by ~20%

Escalation Probability: 0.68

Reasoning:

- Sustained acceleration trend
 - Increasing directional alignment
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Constraints

- No future data leakage
- Simple statistical trends allowed
- Lightweight ML allowed (optional)

- No visualization required
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❷ OPTION 2: Emergency Response Recommendation Engine (Decision-Support Intelligence)

Problem

Even correct risk detection is useless without **clear response guidance**.

Required Logic

For every **HIGH-risk** event, your system must recommend:

1. **Type of response unit**
 2. **Primary action**
 3. **Urgency level**
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Decision Inputs (Must Use at Least 3)

- Risk score
 - Type of anomaly
 - Area affected
 - Duration of risk persistence
 - Escalation trend
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Example Output

Time: 18:44:02

Risk Level: HIGH

Recommended Response:

- Unit: Crowd Control Police
- Action: Restrict entry points, manage flow

- Urgency: Immediate

Reasoning:

- High density + directional chaos

- Risk persistence over multiple intervals

✓ Rule-based decision trees allowed

✓ Hard-coded mappings NOT allowed

📌 OPTION 3: Multi-Zone Risk Propagation & Escalation

(System-Level Intelligence)

Problem

Large venues use **multiple cameras**. Risk rarely stays localized.

Required Logic

Simulate multiple camera zones as **independent data streams**.

Your system must:

- Track risk levels per zone
 - Detect **risk propagation patterns**
 - Escalate risk when anomalies spread
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Mandatory Escalation Rule (Example)

If MEDIUM risk appears in 2 adjacent zones within a short time window → escalate to HIGH.

Example Output

Time: 18:45:30

Zones Affected: Zone 2 → Zone 3

Escalation Trigger:

- Risk propagation detected

Consolidated Risk Level: HIGH

Reasoning:

- Temporal correlation across zones

- Increasing density and motion instability

✓ No video stitching required

✓ Logical correlation only

Explicitly Forbidden

-  UI dashboards
-  Video overlays
-  Face or identity analysis
-  Black-box predictions without explanation

Technical Constraints

- Must extend Day-1 system
- No new datasets required
- CPU-only allowed
- Conceptual simulations allowed
- Ethical constraints remain mandatory

Day-2 Deliverables

- 1** Working extended system
- 2** Sample logs for selected option
- 3** Short explanation covering:
 - Prediction or decision logic
 - Assumptions made

- Limitations acknowledged